

John Doe

Senior Electrical & Controls Engineer

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PROFESSIONAL SUMMARY

Senior Electrical & Controls Engineer with 8+ years of experience leading the design, commissioning, and delivery of control and instrumentation systems across industrial facilities. Proven expertise in DCS, PLC/HMI, and SCADA architecture, with strong capabilities in control philosophy development, P&ID design, and instrumentation standards. Experienced in facility audits, project scoping, cost estimation, and client-facing technical recommendations. Track record of mentoring engineering staff, overseeing multi-discipline project execution, and ensuring compliance with industry safety and regulatory standards.

CORE SKILLS

- Process instrumentation design and specification
- P&ID development and interpretation
- Instrument index and specification sheets (ISA data sheets)
- Loop diagrams and wiring design
- Hook-up drawings and installation details
- DCS configuration (Honeywell Experion, Emerson DeltaV)
- PLC programming (Allen-Bradley, Siemens S7)
- SCADA systems and HMI design
- SIS design per IEC 61511 / ISA 84
- SIL verification and validation
- Instrument calibration and loop checking
- HART / Foundation Fieldbus / Profibus protocols
- Smart transmitter configuration (Emerson, Yokogawa, Endress+Hauser)
- Calibration management (Beamex CMX, Fluke tools)
- Project planning and execution
- Vendor technical evaluation and procurement
- FAT / SAT coordination
- Commissioning and start-up support
- Technical documentation and as-built records
- SmartPlant Instrumentation (SPI / INtools)
- AutoCAD / MicroStation
- MATLAB / Simulink
- SAP PM module
- Microsoft Office Suite
- Functional safety lifecycle management
- HAZOP and LOPA participation
- SIL assessment and classification
- Compliance: IEC 61511, IEC 61508, ISA 84, API 554, NEC/NFPA
- Facility audits and project scoping
- Control philosophy development and architecture design
- Client relationship management and technical advisory
- Tender and construction package preparation
- North American electrical codes (NEC, NFPA 70E, CSA)

PROFESSIONAL EXPERIENCE

Senior Instrumentation Engineer

Jan 2021 - Present

Petronova Engineering Houston, TX

- Lead instrumentation engineering for greenfield and brownfield projects valued at \$50M+
- Develop instrument specifications, data sheets, and hook-up drawings for pressure, temperature, level, and flow instruments
- Design and review P&IDs, loop diagrams, and cable schedules using SmartPlant Instrumentation
- Configure and commission DCS (Honeywell Experion) and SIS (Triconex) for refinery units
- Conduct SIL verification studies and prepare safety requirement specifications (SRS)

- Coordinate with multidisciplinary teams (process, electrical, piping) during FEED and detailed design
- Mentor junior engineers and review deliverables for quality assurance

Instrumentation Engineer

Mar 2017 - Dec 2020

Gulf Coast Process Solutions Pasadena, TX

- Executed instrumentation detailed design for petrochemical and gas processing facilities
- Prepared instrument index, I/O lists, junction box layouts, and cable routing drawings
- Performed loop checking, calibration, and pre-commissioning of 2,000+ instrument loops
- Configured HART smart transmitters and control valves with digital positioners
- Supported DCS (Emerson DeltaV) point-to-point checkout and logic validation
- Participated in HAZOP reviews and implemented LOPA recommendations

Junior Instrumentation Engineer

Jun 2014 - Feb 2017

Meridian Industrial Services Baton Rouge, LA

- Assisted in design and installation of instrumentation for power plant upgrades
- Created instrument hook-up drawings and typical installation details in AutoCAD
- Performed field calibration of pressure, temperature, level, and flow transmitters
- Maintained calibration records and updated instrument database in SPI/INtools
- Supported plant turnaround activities including instrument inspection and replacement

EDUCATION

B.Sc. in Electrical Engineering (Instrumentation & Control)

2014

Louisiana State University Baton Rouge, LA

CERTIFICATIONS

- Certified Functional Safety Engineer (CFSE) - TUV Rheinland
- ISA Certified Control Systems Technician (CCST Level III)
- OSHA 30-Hour General Industry Safety

PROFESSIONAL MEMBERSHIPS

- International Society of Automation (ISA) - Senior Member
- Society of Petroleum Engineers (SPE)

LANGUAGES

- English - Native
- Spanish - Professional working proficiency